

MCELROY AFLD, CO, USA

WMO: 720262

Lat: 40.054N

Lon: 106.369W

Elev: 2259

StdP: 76.97

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94076

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-24.0	-32.1	0.2	-26.6	-27.7	0.4	-22.1	11.2	1.4	9.7	2.1	0.1	90	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	31.4	11.8	29.7	11.7	27.5	11.2	14.7	23.5	14.0	23.1	13.2	22.8	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	11.8	17.3	11.2	10.9	16.3	10.1	10.1	15.5	49.8	24.1	47.0	23.2	44.5	22.2	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.4	7.3	-32.7	34.0	1.7	1.9	-34.0	35.3	-35.0	36.4	-35.9	37.5	-37.2	38.8		
(5)			-32.2	16.0	3.2	1.1	-34.6	16.8	-36.5	17.5	-38.3	18.1	-40.6	18.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-10.3	-5.9	0.2	5.0	9.4	14.8	18.4	16.6	12.6	5.6	-0.4	-8.4	(6)
(7)		DBStd	10.58	6.98	6.97	5.33	3.77	3.61	3.14	2.40	2.20	2.99	4.31	5.21	6.70	(7)
(8)		HDD10.0	2622	628	447	304	152	56	4	0	0	9	141	312	569	(8)
(9)		HDD18.3	4966	887	679	562	399	278	112	29	62	172	393	562	828	(9)
(10)		CDD10.0	744	0	0	1	3	36	147	260	204	88	6	0	0	(10)
(11)		CDD18.3	46	0	0	0	0	0	5	31	8	2	0	0	0	(11)
(12)		CDH23.3	2061	0	0	0	2	47	514	854	477	165	1	0	0	(12)
(13)		CDH26.7	605	0	0	0	0	4	159	308	109	24	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.3	1.9	2.1	2.7	2.3	2.2	1.7	1.6	1.7	1.8	1.6	1.4	(14)
(15)	Precipitation	PrecAvg	342	23	22	22	30	32	29	34	39	43	26	25	16	(15)
(16)		PrecMax	461	58	45	38	54	94	65	61	74	82	48	39	30	(16)
(17)		PrecMin	229	7	8	12	16	8	4	11	13	18	5	14	6	(17)
(18)		PrecStd	61	13	12	7	11	19	15	15	16	18	13	7	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	12.3	19.9	22.6	27.2	32.8	34.8	31.3	29.8	22.8	17.3	10.9	(19)
(20)			MCWB	0.8	4.6	6.3	6.9	9.4	11.0	13.1	12.3	12.1	8.0	5.3	2.5	(20)
(21)		2%	DB	3.6	8.0	16.0	20.0	24.9	31.0	32.3	29.7	27.1	21.2	15.1	6.2	(21)
(22)			MCWB	-0.2	1.9	4.4	6.1	8.8	10.8	12.5	12.0	10.2	7.5	4.3	0.2	(22)
(23)		5%	DB	2.3	6.3	12.5	17.3	22.5	27.8	30.2	27.6	25.2	18.0	12.4	3.0	(23)
(24)			MCWB	-0.6	0.8	3.3	5.0	8.1	10.3	12.3	11.4	9.3	6.5	3.6	-0.7	(24)
(25)		10%	DB	1.0	3.9	9.9	15.0	20.8	26.3	27.7	26.3	22.9	15.8	9.7	2.0	(25)
(26)			MCWB	-1.6	-0.2	2.5	4.3	7.5	9.9	12.0	11.3	9.0	5.6	2.6	-1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	5.3	6.7	7.9	10.9	13.8	16.5	15.4	13.5	10.2	6.3	2.9	(27)
(28)			MCDWB	4.9	11.1	17.4	19.1	23.3	22.3	25.0	22.7	23.1	17.3	15.6	9.2	(28)
(29)		2%	WB	0.7	2.8	5.3	6.7	9.8	12.7	15.3	14.4	12.4	8.6	4.9	1.0	(29)
(30)			MCDWB	2.8	7.4	14.4	17.4	21.4	23.9	24.3	22.4	21.4	17.7	13.5	5.0	(30)
(31)		5%	WB	-0.5	1.6	3.9	5.7	8.9	11.9	14.4	13.6	11.4	7.4	3.8	-0.1	(31)
(32)			MCDWB	1.9	5.8	11.5	15.4	19.9	23.9	23.8	22.1	20.1	16.5	11.4	3.4	(32)
(33)		10%	WB	-1.9	0.2	2.7	4.9	8.0	11.1	13.7	12.8	10.3	6.2	2.6	-1.2	(33)
(34)			MCDWB	0.6	3.9	8.9	13.7	18.1	23.6	23.4	21.7	19.6	14.9	8.8	1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.7	13.6	14.2	15.5	17.1	21.3	20.5	20.0	20.7	17.2	15.6	13.2	(35)
(36)			MCDBR	12.8	13.4	19.0	22.0	22.8	25.0	24.1	22.7	24.3	23.1	20.4	13.3	(36)
(37)			MCWBR	9.7	8.3	9.2	10.0	8.9	8.7	7.6	8.0	9.5	11.1	11.2	9.1	(37)
(38)		5% WB	MCDBR	11.9	12.0	17.5	18.9	20.4	21.3	18.6	17.4	19.0	20.0	18.3	12.2	(38)
(39)			MCWBR	9.4	7.6	8.8	8.9	8.5	8.2	6.6	6.6	7.7	9.9	10.3	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.224	0.236	0.260	0.298	0.310	0.315	0.344	0.340	0.292	0.266	0.241	0.229	(40)	
(41)		taud	2.431	2.375	2.403	2.417	2.490	2.480	2.445	2.438	2.588	2.625	2.579	2.473	(41)	
(42)		Ebn at Noon	974	1002	1005	977	966	956	925	922	958	958	949	941	(42)	
(43)		Edh at Noon	77	96	105	112	107	108	111	109	87	75	67	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.47	3.41	4.82	5.63	6.12	7.32	6.66	5.75	5.08	3.66	2.55	2.04	(44)	
(45)		RadStd	0.12	0.21	0.38	0.41	0.47	0.45	0.32	0.29	0.34	0.31	0.24	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page